File Copy Studio

Update Manual

Version 2026.05.12.0005

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File Copy Studio is a professional folder-to-folder file transfer utility for Windows. Configurable conflict resolution, CRC32 integrity verification, crash-safe resume, built-in job scheduler, and full WCAG 2.2 AAA accessibility - all in a self-contained executable that requires no .NET runtime on the target machine.

1 What's New in v2026.05.12.0005

Version	Highlights
2026.05.12.0005	WCAG AAA: BrushSelectedText PrimaryButton/DangerButton; indeterminate CheckBox visual; BrushRed contrast. Performance: meta flush 16MB to 64MB; single FileInfo stat in Enumerate; CreateDirectory HashSet cache. Scheduler: Save job button contrast; 44px form controls; focus rings on LstJobs. Preset Move-left/right. BtnUiScale AutomationName corrected.
2026.05.07.x	Built-in job scheduler with cron expressions; per-job throttle; tray minimize on scheduled run.
2026.05.04.x	Initial release: copy engine, CRC32, crash-safe resume, parallel workers, presets, job files, shell integration.

1.1 Performance Improvements

* **Resume meta flush** - Checkpoint raised 16MB to 64MB per worker; reduces IOPS on large transfers by ~75%.
* **Directory enumeration** - Single FileInfo stat per entry vs separate Exists+Length+LastWriteTime; ~60% fewer syscalls on network shares.
* **CreateDirectory caching** - HashSet per scan skips redundant Directory.Exists checks for already-created paths.

1.2 Accessibility Fixes

* **BrushSelectedText** - PrimaryButton and DangerButton selected-text colour corrected to >= 7:1 against their backgrounds.
* **Indeterminate CheckBox** - Now uses a filled dash glyph, not grey-on-grey; passes SC 1.4.1 non-colour-only info.
* **BrushRed contrast** - Dark theme BrushRed raised to clear 7:1 AAA against window background.
* **SchedulerWindow** - Save job button contrast corrected; all form controls 44px; focus rings on LstJobs.
* **BtnUiScale AutomationName** - Screen reader now announces current scale level correctly.
* **Preset Move left/right** - Preset slots now support reordering via arrow-button controls.

2 System Requirements

Requirement	Minimum	Recommended
OS	Windows 10 (1809+) x64	Windows 11 x64
.NET runtime	None - self-contained	None - self-contained
RAM	64 MB	256 MB+
Disk	~45 MB (app)	SSD recommended for large transfers
Display	1024x600	1920x1080+

Note: No .NET runtime installation required. The executable is fully self-contained.

3 Update Instructions

3.1 Installer Update

* Close **File Copy Studio** if running (check system tray).
* Download **FileCopyStudio_Setup.exe** from the release page.
* Run installer. Detects existing installation and upgrades in-place.
* All settings, presets, and preferences are preserved.
* On first launch a **What's New** dialog highlights changes in this version.

3.2 Portable Update

* Download **FileCopyStudio_Portable_<version>.zip**.
* Extract to a temporary folder.
* Copy **FileCopyStudio.exe** over the existing portable executable.
* Your **Data** folder (settings, presets, logs) is untouched.

3.3 Silent Update

```
FileCopyStudio_Setup.exe /S
```

Installs or updates silently. Suitable for managed deployment.

3.4 Rollback

Run a previous version's installer to downgrade. Settings are preserved. Versioned builds are in the **Builds** subfolder of the project.

4 Feature Overview

Feature	Detail
Conflict resolution	Overwrite, Overwrite-if-newer, Overwrite-if-older, Skip-if-same-date, Skip-all, Rename.
CRC32 verification	Post-copy verify; standalone Verify Only mode; .crc sidecar option.
Crash-safe resume	.fctemp + .fctemp.meta sidecar; 64 MB checkpoint; auto-detected on next scan.
Parallel workers	Auto-detects same/different volumes; configurable 1-8 threads.
Bandwidth throttle	Configurable cap in Mbps per session or per scheduled job.
Dry-run mode	Plans actions, shows file grid, writes nothing to disk.
Symlink policy	Follow, Skip, or Copy-as-link.
File masks	Include/exclude semicolon-separated wildcards (*.jpg;*.png).
Built-in scheduler	Cron-style expressions; per-job throttle and conflict options.

Feature	Detail
5 named presets	Store paths + all options; Ctrl+1-5 to load; Ctrl+Shift+1-5 to save.
Job files (.fclist)	JSON format; repeatable batch transfers; shell context-menu integration.
Portable mode	portable.flag next to exe; prefs+logs in Data\; no registry; USB-ready.

5 Accessibility (WCAG 2.2 AAA)

This application targets WCAG 2.2 Level AAA across all interactive surfaces.

Criterion	Implementation
Contrast (SC 1.4.6 AAA 7:1)	All text tokens >= 7:1 on Dark and Light themes.
Non-text contrast (SC 1.4.11)	All icons, borders, focus indicators >= 3:1.
Touch targets (SC 2.5.5 AAA 44px)	All interactive controls MinHeight=44 MinWidth=44.
Focus appearance (SC 2.4.11 AAA)	2px Gold (dark) / Cyan (light) focus ring on all controls.
No colour-only info (SC 1.4.1)	Status icons use glyphs alongside colour coding.
Live regions (SC 4.1.3)	Status text and progress exposed via AutomationPeer.
High-contrast mode	Detects Windows High Contrast; swaps to system palette.
UI scale	Five scale levels (Compact to Extra Large).

6 Known Issues & Notes

Scan returns 0 files

Verify source path exists and is accessible. Check include mask (default *.*)

Files fail CRC verification

Indicates read or write error. Retry the copy. Persistent failures suggest media errors.

Scheduled job does not fire

Application must be running or in system tray. Check cron expression and log file.

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